Discrete Drainage Dynamics: A Bounded, Conservative Local Rule for Newtonian-Like Potentials

Stanislas Dewavrin[1]

[1]Independent researcher , dewavrin.iphone@gmail.com

April 2026

Abstract

Discrete Drainage Dynamics (DDD) is an analogue-substrate programme. It does not contest General Relativity or the Standard Model: those are treated as the empirical anchors that any analogue must reproduce in the appropriate continuum limit. DDD asks the more modest, exploratory question of whether some of the *forms* that appear fundamental in current physics can be made mechanically legible inside a deliberately simpler, local, discrete, finite-reserve substrate. The programme is exploratory and conditional; the present paper introduces the substrate.

We define a deterministic, parallel, local update rule on a discrete graph, constrained by six operational axioms (locality, determinism, internal conservation, reserve non-negativity, finite per-link bandwidth, large-scale isotropy). Each node carries a non-negative reserve $R_i \geq 0$ with rest value R_0; each oriented link carries a non-negative directional flux; the rule alternates a link update and a node update with a per-node rate-limiter $\beta_i \in [0, 1]$ that prevents the reserve from becoming negative. The flux on a link is written in a general form involving a *mobility function* $\mu(R)$, the choice of which is left as an explicit hypothesis of the framework (companion papers explore alternative hypotheses). The present paper adopts the constant-mobility hypothesis $\mu \equiv 1$ as the calibration regime in which finite-resource non-linearity is inactive. In this regime, far from the non-negativity floor, the rule's steady state coincides with the discrete Poisson equation, and we measure on the lattice that the static profile around a point source follows a $1/r$ form with the continuum coefficient at the percent level. We emphasise that the $1/r$ form is the *expected* analogue baseline (any 3D discrete diffusion produces it); the contribution of this paper is the bounded-reserve, rate-limited rule and the mobility hypothesis framing, which together provide a finite-resource analogue substrate for the non-linear phenomena studied in companion papers. We do not derive Newton's G, GR, or any aspect of the Standard Model; we calibrate G from observation and treat the underlying physical theories as anchors, not as targets to replace.

What this paper makes mechanically legible. Newtonian gravity treats finite signal-propagation speed as an empirical input (deferred to GR's postulate of c) and a plain Poisson equation has no mechanism preventing pathological sinks. *This paper* introduces a discrete substrate whose locality (L1) and finite per-link bandwidth (L5) *force* a propagation-speed ceiling by construction, and whose non-negativity rate-limiter (H-ratelimit) regulates would-be singularities at the substrate level. **Leverage:** the existence of a propagation ceiling and the regularity of sources become *consequences* of the discrete rule, not postulates — the ontological seed for Papers II–IV's mechanical readings of g_{tt}, c, and the deflection factor of two.

1 Programme positioning and analogue framing

Before stating the claims of this paper, we make explicit the position the entire DDD programme takes with respect to established physics, because misreading on this point would collapse the rest of the work into something it does not attempt to be.

What DDD is. DDD is an analogue-substrate programme. It explores whether some *forms* that appear fundamental in continuum physics (e.g. a static $1/r$ potential, a Schwarzschild-like clock-rate factor, a Lorentz-like dispersion, gauge-like couplings) can arise as the continuum limit of simple local update rules on a finite-reserve discrete substrate. The programme is exploratory: it identifies which simple substrate assumptions are sufficient to reproduce selected continuum forms, where those analogies break down, and whether remaining unexplained constants tie to discrete structural features.

What DDD is not. DDD does not contest General Relativity, Newtonian gravity, or the Standard Model. Those empirical theories are treated as anchors the analogue must reproduce in the appropriate continuum limit; any deviation between the analogue and the empirical theories is a falsifier of the analogue, not of the empirical theories. The programme does not aim to overthrow the current formalism; it aims to make some of its most mysterious structures *mechanically readable* inside a simpler model. We use phrases such as "recover the form of", "reproduce the analogue of", "substrate-internal derivation" rather than "derive" unconditionally; bare statements of the form "we derive [law of physics]" are out of scope and never claimed.

Empirical anchors and falsifiability. Throughout the series, GR and the SM provide the empirical anchors. The analogue is held to the standard of (i) reproducing the appropriate continuum form within a stated regime of validity, (ii) being explicit about where it breaks down (UV corrections, finite-size effects, scope limits), and (iii) being falsifiable by precision tests of the relevant continuum prediction. The reader should regard each "derivation" in this and subsequent papers as internal to the analogue, conditional on stated hypotheses, and matched against the empirical anchor as a consistency check rather than a fundamental claim.

Which mystery does this paper address? Per the targeted mystery list of the DDD programme (SERIES_FEEDBACK §0.bis), this paper contributes to two *why*-questions, in foundational form:

- **Mystery #1 (the speed-of-light ceiling).** The finite per-link bandwidth α and the discrete tick τ together produce a propagation-speed ceiling $\sim \alpha\,\ell$ in the analogue. The present paper does not yet identify this with the empirical value of c; that identification awaits Paper III's Floquet sector.
- **Mystery #2 (gravitational potentials).** The constant-mobility baseline reproduces the form of Newton's $1/r$ as the analogue's weak-field static limit. The present paper does not address why the analogue's coefficient matches the empirical value of G; that calibration is the subject of Paper II.

The contribution is incremental: this paper supplies the analogue substrate within which subsequent papers address the quantitative *why*-questions for c, G, gravitational time dilation, kinematic time dilation, etc.

Distinctive contribution and ontological reading. The $1/r$ form is not novel as a mathematical result — any 3D discrete diffusion produces it, and Newtonian gravity describes it precisely. The contribution of this paper is therefore *not* the $1/r$ law itself but the substrate *ontology* from which it arises and the structural features that distinguish a finite-bandwidth lattice rule from a plain continuum diffusion. Three distinctive ingredients are introduced, each absent or implicit in the standard treatment:

1. **Finite per-link bandwidth.** The propagation coefficient α is a structural rate, not a free phenomenological parameter; it caps how fast information can move on the substrate. Standard Newtonian gravity has no mechanical statement of *why* signals propagate at finite speed at all (this question is deferred to GR's postulate of c, which itself is an empirical input). DDD's locality axiom forces a propagation-speed ceiling $\sim \alpha \ell$ *by construction*, before any identification with c. This is the foundational ingredient that Paper III uses to read the speed-of-light ceiling as a substrate consequence.

2. **Reserve non-negativity floor and the rate-limiter (H-ratelimit).** The discrete rate-limiter $\beta_i \in [0, 1]$ enforces $R_i \geq 0$ at the substrate level. In the continuum analogue, this floor would not exist: a plain Poisson equation has no mechanism preventing unbounded sinks or pathological singularities. Here, the rate-limiter is a structural regulator that anticipates the *singularity-resolution* programme of Paper II: where GR predicts a curvature singularity at $r = 0$ (the marker of the continuum theory's limit), the discrete substrate has a non-negativity floor that activates before the analogue's $R \to 0$ horizon and prevents the pathological divergence.

3. **Two-step bipartite alternation (H-step-order).** The Link/Node alternation is locally causal by construction: each sub-step reads a fixed neighbouring state and writes the corresponding update, with no self-reference within a tick. In the continuum, causality is imposed via Lorentzian metric structure post-hoc; here it is built into the substrate's update topology from the outset.

Why this matters for the rest of the series. The substrate ontology of this paper is the foundation on which all subsequent papers stand. Paper II uses the finite-reserve floor to read Schwarzschild g_{tt} as bandwidth throttling *and* to begin regularising the BH singularity. Paper III uses the finite per-tick bandwidth to read c as a substrate consequence rather than an ad-hoc postulate. Paper IV uses the same bandwidth profile to derive the weak-field deflection coefficient. None of these moves would be possible in a continuum theory without the discrete substrate's structural ingredients (i)–(iii). The present paper is not a result paper: it is the *ontological seed* the rest of the series grows from.

2 Foundational principles of the analogue substrate

We constrain the substrate by six operational axioms. They are operational in the sense that each is a statement about what the local update rule may or may not depend on, not a statement about deeper physics.

(L1) **Locality.** Every elementary update reads only a node and its immediate neighbours through incident links. There is no action at a distance within a tick.

(L2) **Determinism.** The rule is fully specified by the current substrate state; no randomness or hidden variables enter the elementary update.

(L3) **Internal conservation.** Inter-node flux exchange preserves total reserve exactly. External sinks (e.g. a point source) enter as explicit bookkeeping at the boundary, not as an internal violation.

(L4) **Reserve non-negativity floor.** The reserve satisfies $R_i \geq 0$ at all times: a node cannot push reserve it does not have. A per-node rate-limiter $\beta_i \in [0, 1]$ (Eq. (4) below) enforces this without flux clipping at the link level.

(L5) **Finite per-link bandwidth.** Outgoing flux on a link is bounded; propagation is time-stepped, not instantaneous. The basic propagation coefficient α has the dimensions of a rate.

(L6) **Large-scale isotropy.** The continuum limit of the rule is rotationally symmetric. We work on regular graphs (cubic, BCC) of degree ≥ 6 on which this is realised.

Class constrained by (L1)–(L6). Within (L1)–(L6), the admissible local rules form a class of two-step bipartite alternations (link-then-node updates) on regular lattices with a per-node rate-limiter; we adopt the minimal linear-flux representative described in Sec. **??**. The bipartite-alternation structure is not *forced* by (L1)–(L6) alone — a fully synchronous update is not strictly excluded, but it would require a precedence convention that effectively reintroduces an alternation; we adopt the explicit two-step form for clarity and to match Paper III's stateful-link extension.

Hypothesis (H-mobility): the mobility function is a structural choice. Within the (L1)–(L6) class, the form of the mobility function $\mu(R)$ is not uniquely determined: the axioms admit a family of source-side modulations parameterised by μ. We therefore declare $\mu(R)$ a *hypothesis* of the framework, not a derivation:

(H-mobility) The directional flux on a link is modulated by a non-negative mobility function $\mu(R)$ of the source node's reserve, normalised so that $\mu(R_0) = 1$ at rest. The choice of μ is a structural hypothesis whose microscopic origin is not addressed in this paper; companion papers (notably Paper II) study alternative choices.

Hypothesis (H-const): constant-mobility baseline. The present paper adopts:

(H-const) $\mu(R) \equiv 1$. This models a regime in which source-side bandwidth throttling is inactive. It is the *calibration regime* in which the rule reduces, in the linearised continuum limit, to the discrete Poisson equation and recovers the analogue $1/r$ baseline. (H-const) is a sub-hypothesis of (H-mobility); alternative choices, such as $\mu(R) = R/R_0$ studied in Paper II, define qualitatively different non-linear regimes.

The rule, the rate-limiter, and the linearised-continuum analysis below are therefore conditional on (L1)–(L6) + (H-mobility) + (H-const). The microscopic origin of (H-mobility) is the principal open problem of the analogue framework (Sec. 11).

3 Main claims and scope

We make three claims.

1. We define a deterministic, parallel, two-step local rule on a discrete graph. The reserve is bounded below by zero ($R_i \geq 0$, the physical floor: one cannot drain what is not there); R_0 is the rest value, but the reserve is free to exceed it in excited regimes treated in companion papers. The flux on a link involves a mobility function $\mu(R)$ that we leave as a structural choice of the framework; the present paper studies the constant-mobility baseline $\mu \equiv 1$.

2. Run as a stationary linearised simulation with a point source on a cubic lattice, the rule (with $\mu \equiv 1$) produces a radial profile compatible with $1/r$ in the bulk, with a coefficient that matches the discrete continuum prediction at the percent level across $L \in \{64, 128, 160\}$.

3. The internal flux-exchange rule preserves total reserve exactly. Non-negativity of the reserve is enforced by rate-limiting the outgoing fluxes and external sinks of a node in proportion, so that emission stops when the reserve reaches zero; no mass is created or destroyed by the internal rule. This is the structural feature that distinguishes the rule from a plain discrete heat equation.

What we do *not* claim (defensive restatement). The constant-mobility analogue baseline of this paper is not a claim about fundamental physics. We do not derive Newton's gravitational constant in physical units; we calibrate it against the observed value. We do not derive General Relativity, the Newtonian limit of GR as a fundamental law, or any property of the Standard Model: those theories are empirical anchors, treated outside the analogue. We do not show that the rule is unique. We do not derive the mobility function $\mu(R)$; the constant-mobility baseline of this paper is a choice within a family of possibilities. We do not assert that the $1/r$ result is itself novel: it is not. What is novel is the rule, the mobility framing, and the structural features (reserve non-negativity, rest level R_0, two-step alternation, rate-limited fluxes) that make the rule non-trivial in the regimes explored in the companion papers.

Motivation and significance

The purpose of this paper is *not* to rediscover the $1/r$ potential. Any linear diffusion or graph-Laplacian model in three dimensions has a Green function with this behaviour. The purpose is instead to define a local, conservative substrate rule whose weak-field limit reduces to the discrete Poisson equation, while retaining *finite-resource* structure outside that limit.

This distinction matters for the rest of the DDD programme. A plain heat equation can reproduce $1/r$, but it has no physical floor, no finite local budget, no source-side mobility, and no natural saturation regime. The rule introduced here adds these ingredients: every node carries a non-negative reserve $R \geq 0$, every outgoing flux is rate-limited by the available local budget, and the mobility $\mu(R)$ controls how the source node's reserve modulates emission. These are the substrate features on which the companion papers depend.

The baseline $\mu \equiv 1$ studied in this paper is therefore a *calibration and validation regime*. It verifies that the rule reduces to the expected Poisson behaviour when the finite-resource constraint is inactive. The non-trivial physics begins when $\mu(R) \neq 1$ (Paper II's clock-rate suppression and Schwarzschild geometry), or when the reserve floor $R = 0$ becomes active (saturation, horizon-like bandwidth limit), or when the spinor and gauge-sector back-reactions of Papers III, VI, and VII activate. Paper I supplies the conservative local substrate on which they all depend.

A diagnostic of this finite-resource structure is that superposition of source profiles, exact in the linear regime, must *break down locally* when the reserve floor or the mobility nonlinearity becomes active. We exhibit this directly in §8.4 as a sanity check.

4 Physical picture in plain words

Imagine a fine three-dimensional grid. At each grid point sits a reservoir holding a non-negative amount of stuff. There is a rest level R_0 that the reservoirs all sit at when the system is undisturbed; reservoirs can be excited above R_0 or drained below it, but they cannot go below zero. Between neighbours run pipes that carry a non-negative flow of stuff from the fuller reservoir to the emptier one, at a rate proportional to the level difference, optionally modulated by how much stuff the source actually holds. A reservoir cannot deliver more stuff than it actually has: a node at zero stops emitting.

If we start the system at the rest level R_0 everywhere, drain one point, and let the system settle, the bulk reservoir levels relax into a stationary pattern. With the simplest choice of mobility ($\mu \equiv 1$, the case of this paper), far from the source the pattern reduces to the textbook discrete Poisson solution and the deficit $R_0 - R_i$ falls as $1/r$ with distance. Other choices of mobility yield qualitatively different non-linear regimes, studied in companion papers.

The interesting cases are where the floor matters. Near a strong drainage point, the reservoir level can be driven to zero, and emission from that node stops. In this regime the rule is no longer a discrete heat equation: it is a non-linear update with a finite-resource bound. The companion papers explore those regimes (Schwarzschild-form behaviour in Paper II, Floquet spinor sector in Paper III, gauge sector in Paper VI, gravity-gauge coupling in Paper VII). The present paper is restricted to the linearised regime with $\mu \equiv 1$, where the rule reproduces $1/r$.

5 The substrate

5.1 Graph and variables

The substrate is a graph $G = (V, E)$ in which every node has at least six neighbours and the connectivity is statistically isotropic at large scale. We do not specify the exact topology; in our simulations we use a regular cubic lattice for clarity, and we have verified that BCC gives the same large-scale spectral dimension.

For the purposes of this paper, each node $i \in V$ carries a non-negative reserve $R_i \geq 0$, with rest value R_0 at equilibrium. Each oriented link $(i, j) \in E$ carries a pair of non-negative directional reserve fluxes $F_{i\to j} \geq 0$ and $F_{j\to i} \geq 0$. These are the only variables that enter the linearised gravitational simulation reported here. The non-negativity $R_i \geq 0$ is the physical floor (one cannot drain what is not there).

Variables of the wider rule

The full rule, as used in the companion papers, also assigns to each node two angular quadratures (T_i, I_i) subject to a local coherence bound $T_i^2 + I_i^2 \leq R_i$, and to each oriented link a phase $\psi_{ij} \in [-\pi, \pi)$ with $\psi_{ji} = -\psi_{ij}$. These additional variables drive the gauge sector (Papers V–VI), couple to the reserve through the coherence bound, and do not enter the static linearised gravitational regime treated below.

5.2 Parameters

Parameter	Role	Used in this paper
κ	drainage coefficient	yes
α	flux propagation coefficient	yes
R_0	reserve rest value (equilibrium)	yes
$\mu(R)$	mobility function (structural choice)	yes ($\mu \equiv 1$)
J	link relaxation coupling	no (gauge sector, deferred)

Table 1: Parameters of the local rule. The reserve floor at 0 is not a parameter: it is the physical statement that one cannot drain what is not there. R_0 is the rest level at equilibrium, not an absolute ceiling: the reserve can exceed R_0 in excited regimes treated in the companion papers. The mobility $\mu(R)$ is a structural choice of the framework: this paper uses $\mu \equiv 1$, companion papers (notably Paper II) use other choices.

6 The two-step tick

Motivation

The class of rules considered here is motivated by an old line of inquiry on whether physics admits a microscopic deterministic, local, iterative description — a "computational substrate" in the sense of Zuse, Fredkin, 't Hooft, Wolfram, and Lloyd [1–5]. We do not commit to the strong form of this hypothesis. We adopt it as a working stance.

The rule

The rule decomposes into two alternating half-steps, which we call *Link Update* and *Node Update*. The bipartite alternation is not strictly forced by axioms (L1)–(L6) alone: a synchronous single-step update on stateless links would also satisfy them. We therefore declare it a structural hypothesis:

(H-step-order) The rule is bipartite, with Link Update preceding Node Update within each tick. The bipartite structure is motivated by (i) any synchronous update that simultaneously reads and writes a node and its incident links requiring a precedence convention logically equivalent to the alternation; (ii) the bipartite form preserving locality (L1) and avoiding self-reference within a tick by construction; and (iii) the match with the stateful-link extension of Paper III, where the alternation becomes *necessary* because the link variables are dynamical. The Link-then-Node ordering (rather than the reverse) is a convention; the reverse ordering would also satisfy (L1)–(L6) and lead to a related class of rules. (H-step-order) is therefore conditional on the same operational considerations as (H-mobility); its microscopic origin is not addressed here.

Constraint-class argument for the two-step bipartite alternation. Within axioms (L1)–(L6), we can be more precise about *why* the two-step bipartite alternation is the canonical representative of the rule class:

1. By (L1) locality, an elementary update at node i depends only on $\{R_i, R_{j\in N(i)}\}$ and the incident link states.
2. By (L2) determinism, the update is a fixed function of its inputs.
3. By (L3) conservation, the per-tick update preserves $\sum_i R_i$ exactly (modulo external sinks).
4. Any update that simultaneously reads and writes a node i and its incident link (i, j) in the same elementary step requires a precedence convention to resolve which value feeds which. Adopting that convention is logically equivalent to splitting the elementary step into two halves — one that reads the link state into the node, one that reads the node state into the link. This is exactly the bipartite alternation.
5. Conversely, any non-alternating decomposition either allows a freshly-rewritten variable to feed the same step (violating no-self-reference within the step) or duplicates state by buffering both old and new values (the alternation in disguise).

The two-step structure is therefore the *minimal non-trivial sequentially-causal decomposition* compatible with (L1)–(L3) on a graph where both nodes and links carry independent state. The Link-then-Node ordering (rather than Node-then-Link) is the only convention not fixed by the axioms, and is the residual content of (H-step-order). This argument generalises to Paper III's stateful-link extension where the alternation becomes *necessary* because link variables are dynamical.

(remaining open) None of (L1)–(L6) require the rule to be *first-order* in time on the scalar reserve sector; that further property is what enforces the diffusive character of Paper I's $\mu \equiv 1$ baseline. Future Floquet or oscillator extensions (Paper III's spinor sector) relax that first-order property at the spinor locus only.

6.1 Link Update

During the first half-tick, the node reserves are frozen. For each oriented link (i, j), in parallel, we compute a desired flux. We write the desired flux in the general form

$$F^{\rm des}_{i\to j} \;=\; \alpha\,\mu(R_i)\,(R_i - R_j)_+, \tag{1}$$

where $\mu(R)$ is a non-negative *mobility function* encoding how the available reserve at the source node limits its ability to push reserve into a neighbour. We normalise $\mu(R_0) = 1$ at the rest level, so that α retains its meaning as the rated propagation coefficient when the source is at rest. We do not derive $\mu(R)$ from a deeper principle; it is a structural choice of the rule.

Constant-mobility baseline of this paper ((H-const), Sec. 2). The present paper adopts the constant-mobility hypothesis,

$$\mu(R) \;\equiv\; 1, \tag{2}$$

for which Eq. (1) reduces to the textbook discrete-diffusion form $F^{\rm des}_{i\to j} = \alpha(R_i - R_j)_+$, and the linearised steady state is the discrete Poisson equation, giving the $1/r$ profile measured in Sec. 8.

Other mobility choices in companion papers. The reserve-proportional mobility $\mu(R) = R/R_0$, motivated by local bandwidth limitation at the source node, is studied in Paper II, where it is shown to produce $\nabla^2(R^2) = 0$ in the continuum limit and a profile of Schwarzschild form. Other mobility prescriptions may be explored in subsequent papers. Within Paper I, none of these is invoked.

The two directions of a link are independent. Each node then computes a rate-limiting factor that ensures the reserve stays non-negative on its outgoing side. Define

$$A_i \;=\; R_i \,+\, \tau\, S^{\rm in}_i, \quad S^{\rm in}_i = \sum_j F^{\rm des}_{j\to i}, \quad D_i \;=\; \tau\, O^{\rm des}_i + \tau\,\kappa\, E_i, \quad O^{\rm des}_i = \sum_j F^{\rm des}_{i\to j}. \tag{3}$$

The rate-limiter is

$$\beta_i \;=\; \begin{cases} 1, & D_i = 0, \\ \min(1,\; A_i/D_i)\,, & D_i > 0, \end{cases} \tag{4}$$

(H-ratelimit) Per-node balanced rate-limiter as minimal enforcer of (L4). Equation (4) is not the unique enforcement of the non-negativity floor (L4): one could instead clip individual link fluxes, use an adaptive sub-tick step, or apply a softening function. We adopt the per-node balanced choice as a structural hypothesis because (i) it scales *all* outflows and external sinks at node i identically, preserving the proportion of demand met; (ii) it activates only when demand exceeds available reserve, leaving the rule untouched in the linear regime; and (iii) it is the minimal enforcement compatible with internal conservation (L3). Alternatives are not explored in this paper; (H-ratelimit) is on the same logical footing as (H-mobility) and (H-step-order).

The rate-limiter is applied identically to outflows and sink:

$$F_{i\to j} \;=\; \beta_i\, F^{\rm des}_{i\to j}, \qquad E^{\rm real}_i \;=\; \beta_i\, E_i. \tag{5}$$

6.2 Node Update

During the second half-tick, the link fluxes are frozen.

$$R_i \;\leftarrow\; R_i + \tau\, S_i^{\rm in} \;-\; \beta_i\, \tau\, O_i^{\rm des} \;-\; \beta_i\, \tau\, \kappa\, E_i. \tag{6}$$

By construction of β_i, this update keeps R_i non-negative at every tick.

7 Conservation

Total reserve, exactly conserved by internal exchange. The realised fluxes enter the Node Update of j as $+\beta_i F^{\rm des}_{i\to j}$ and the Node Update of i as $-\beta_i F^{\rm des}_{i\to j}$. Summed over the graph, $\sum_i R_i$ is exactly conserved by inter-node flux dynamics. The rate-limiter scales outflows and the local sink in the same proportion.

External sinks as bookkeeping. The total reserve in the network changes only by $\sum_i E_i^{\rm real}$. The closed system network-plus-external-sink is conservative; the network alone is open.

8 Simulation: from rule to $1/r$

8.1 Derivation of the Poisson limit (constant mobility)

We work in the constant-mobility baseline $\mu \equiv 1$. In the linear regime $\beta_i = 1$, the two directional fluxes satisfy $F_{i\to j} - F_{j\to i} = \alpha(R_i - R_j)$, so $S_i = \alpha\Delta R_i$ where Δ is the discrete graph Laplacian. In stationary regime, $\alpha\Delta R_i = \kappa E_i$. Switching to the deficit $\delta_i = R_0 - R_i$:

$$-\Delta\, \delta_i \;=\; M\, \delta_{\rm Kr}(i, i_0), \qquad M \;=\; \frac{\kappa E_0}{\alpha}. \tag{7}$$

The continuum solution is $\delta(r) = M/(4\pi r)$.

8.2 Numerical check

L	$A_{\rm fit}$	relative bias	statistical precision	$N_{\rm iter}$
64	0.07878	-1.01%	7.0×10^{-5}	5 000
128	0.07866	-1.16%	5.7×10^{-6}	10 000
160	0.07886	-0.90%	3.7×10^{-6}	30 000
$A_{\rm continuum} = M/(4\pi)$		0.07958		

Table 2: Newton-fit coefficients on cubic lattices, $M = 1$, Dirichlet boundary, with $\mu \equiv 1$. The bias is roughly L-independent at the percent level (Madelung correction).

8.3 Robustness: spectral dimension

8.4 Two-source superposition and onset of finite-resource effects

As a second sanity check that distinguishes the DDD rule from a plain graph-Laplacian, we place two identical point sinks at separation $d = 8$ lattice spacings and compare the steady-state deficit field δ_{12} with the linear superposition $\delta_1 + \delta_2$ obtained from two independent one-source runs.

Linear regime ($\mu = 1$, weak sinks). For sink strength 0.001, the reserve floor is never approached ($R_{\rm min} \simeq R_0 - \mathcal{O}(10^{-4})$, with $R_{\rm min}/R_0 > 0.999$ everywhere), the rate-limiter $\beta_i \equiv 1$ throughout, and the rule reduces to the discrete Poisson equation. The relative residual

L	cubic ($d_s - 3$ minimum)	BCC ($d_s - 3$ minimum)
32	0.129	0.107
64	0.054	0.045
128	0.023	0.019

Table 3: Spectral-dimension residual converges to 0 in the IR limit on both substrates. Newtonian gravity recovered. This convergence validates the large-scale isotropy axiom (L6).

$\max|\delta_{12} - (\delta_1 + \delta_2)| / \max \delta_{12}$ is 5.4×10^{-3}, attributable entirely to lattice and boundary truncation. Superposition holds.

Saturating regime (rate-limiter active). For sink strength 12, the reserve at each source is driven down to the floor $R = 0$. The rate-limiter $\beta_i = \min(1, A_i/D_i)$ (where A_i is the available reserve and D_i is the demand from outgoing fluxes plus external sinks at node i) drops below 1 at depleted nodes, capping the outgoing flux, and the steady-state field no longer obeys superposition: the relative residual rises to 2.7×10^{-2}, an order of magnitude above the lattice-error baseline, and is concentrated near the saturated sources where δ_1 and δ_2 would otherwise have added linearly.

This contrast — $\sim 0.5\%$ residual when the floor is inactive vs $\sim 3\%$ when it is active — is the diagnostic signature of finite-resource physics that motivates the rule. The deviation is not a numerical defect; it is the substrate correctly refusing to deliver more flux than its local reserve permits. Figure 1 displays the central slice of both regimes side-by-side.

8.5 Code and data availability

Scripts available at `https://github.com/stanislasdewavrin/DDD-Paper-1-Foundations`. Pure-numpy, no scipy.

9 Calibration vs. prediction

Calibration. The mappings $\delta \leftrightarrow \Phi$ and $E \leftrightarrow m$ are external. The lattice unit and κ are fixed by demanding A_{fit} matches Newton's G.

Prediction. Within that family, the rule (with $\mu \equiv 1$) predicts that the static potential is $1/r$ on the linearised regime.

10 What the model is not, and what it does not do

We do not derive general relativity, Newton's G, the speed of light c, or any property of the Standard Model.

We do not specify the exact graph topology beyond "minimum 6 neighbours, statistically isotropic". We have tested cubic and BCC.

We do not derive the mobility function $\mu(R)$. The present paper studies the constant-mobility baseline $\mu \equiv 1$, for which the linearised steady state is the discrete Poisson equation. Different mobility choices lead to different non-linear steady-state equations; the constant-mobility choice is the minimal Newtonian baseline. Companion papers study specific reserve-dependent mobility prescriptions, in particular the source-side bandwidth-throttled choice $\mu(R) = R/R_0$ in Paper II.

We have not shown that the rule is unique.

Calibration ledger (per SERIES_FEEDBACK §0.ter). The present paper introduces the analogue substrate. The following table lists each quantity introduced and its status:

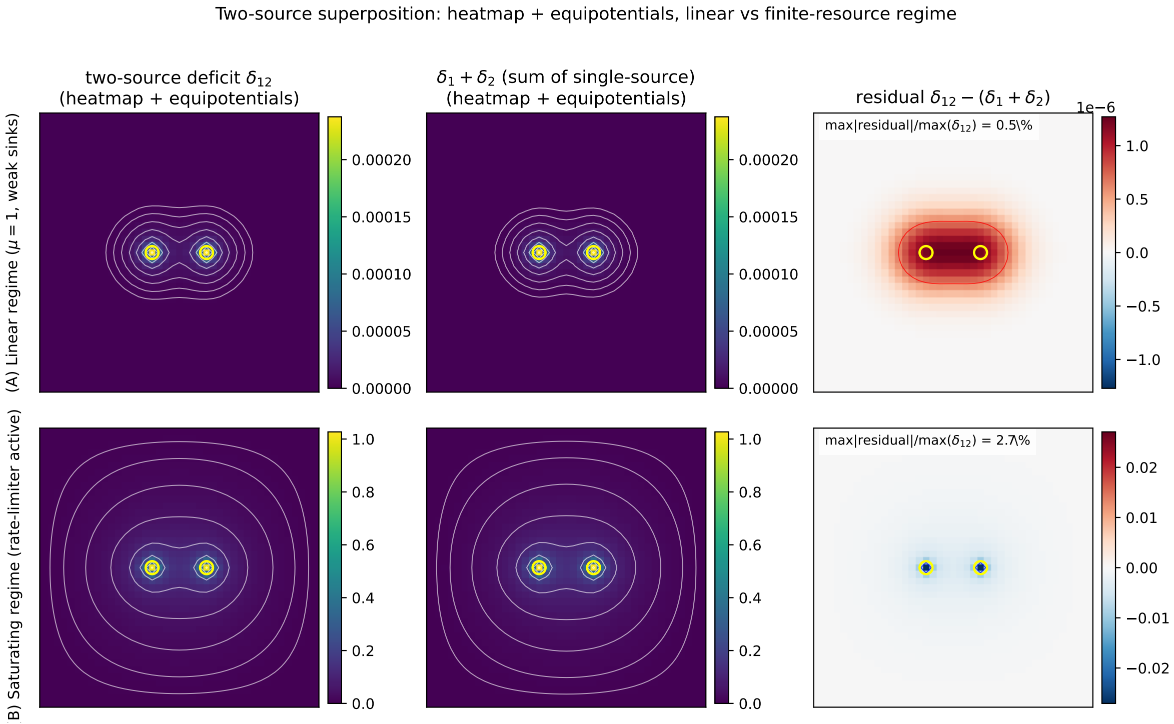

Figure 1: Two-source superposition test. **Top row (A)**, linear regime ($\mu = 1$, weak sinks): the two-source deficit δ_{12} (left), the superposition $\delta_1 + \delta_2$ (centre), and the residual (right). The residual is at the 0.5% level and consists entirely of lattice/boundary error. **Bottom row (B)**, saturating regime (rate-limiter active): the residual rises to 2.7% and is localised at the sources where the reserve floor caps the outgoing flux. *Heatmap + equipotentials*: white contour lines mark log-spaced equipotentials of the deficit field; on the residual panel, the black contour is the zero line and the red/blue contours mark ±50% of the residual maximum. Source positions marked with yellow circles. Plane shown is $z = z_{\text{centre}}$. Colourmaps: viridis for absolute deficit, RdBu for the residual (symmetric about zero). Reproducible via `code/11_two_source_superposition.py`.

Quantity	Status	Calibration source / future resolution
$R_i \geq 0$ (reserve)	substrate field	defined; not calibrated
R_0 (rest level)	convention	arbitrary unit; sets dimensional scale
$F^{\text{des}}_{i\to j}$ (flux)	substrate field	defined; not calibrated
α (propagation rate)	calibrated	combined into Newton's G in Paper II via Planck-anchoring (CODATA G)
τ (discrete tick)	calibrated	combined into the speed-ceiling identification in Paper III
$\mu \equiv 1$ (constant-mobility baseline)	hypothesis (H-const)	sub-hypothesis of (H-mobility); future origin of mobility law open
β_i rate-limiter form	hypothesis (H-ratelimit)	no derivation; alternative limiters not explored
two-step bipartite alternation	hypothesis (H-step-order)	convention; reverse order would also satisfy (L1)–(L6)
$1/r$ steady-state form	derived	from (L1)–(L6) + (H-const) in continuum limit; not specific to DDD

Rule loci touched and no-epicycle statement (per SERIES_FEEDBACK §0.quater). The present paper introduces content at the following loci of the analogue substrate, and at no other:

- *Nodes:* per-node reserve $R_i \geq 0$.
- *Links:* per-link directional flux $F_{i\to j} \geq 0$.
- *Update rule:* two-step bipartite alternation (Link Update then Node Update); flux equation Eq. (1) with constant-mobility baseline $\mu \equiv 1$; per-node rate-limiter $\beta_i = \min(1, A_i/D_i)$.
- *Substrate:* regular cubic graph (and BCC variant for the spectral-dimension test), degree ≥ 6.
- *Configuration:* a single point sink in static linearised conditions.

No-epicycle statement. No additional fields, no extra free parameters, no relaxation of the principles (L1)–(L6) of Sec. 2 have been silently introduced. Every quantity used in the paper is either listed in the calibration ledger above or is a direct consequence of the rule applied to the configuration. Future papers extending the analogue (Paper II adds $\mu(R) = R/R_0$ on the update locus; Paper III adds spinor on nodes and twist on links; Paper IV adds photon test particles on configuration) do so by labelled extensions at named loci, never by silent addition.

11 Open questions

Why six neighbours? Six is the minimal cubic coordination; not a fundamental requirement. More general graphs require separate checks of spectral dimension and isotropy.

Why this order, Node then Link? Microscopic breaking of time-reversal symmetry; we have no derivation.

What sets R_0? The only structural scale of the rule. Not derived.

What sets the mobility $\mu(R)$? The constant-mobility baseline $\mu \equiv 1$ used here is the simplest choice. Other choices (in particular the source-side bandwidth-throttled $\mu(R) = R/R_0$ studied in Paper II) yield qualitatively different non-linear regimes. A microscopic argument that selects a specific mobility from the rule would close an important gap of the framework.

Is the rule unique? See Sec. 10.

What sits below the substrate? Beyond the present paper.

12 Outlook

1. **Paper II** — the source-side reserve-proportional mobility $\mu(R) = R/R_0$ yields a Schwarzschild-like static clock-rate factor in the spherical continuum limit, recovering weak-field general-relativistic time dilation from the local substrate rule. *Conditional on the mobility choice and on the bandwidth-time identification.*
2. **Paper III** — a Floquet two-step spinor sector on the same substrate supports propagating localised excitations and an operational, kinematic Lorentz-like clock-rate.
3. **Paper IV** — calibrates the photon-substrate coupling against the static spherical sector and reproduces light deflection and Shapiro delay at the percent level.

4. **Paper V** — collects candidate predictions and falsification windows of the gravitational sector (Yukawa bounds, short-range tests, feedback-mediated effects).

5. **Paper VI** — extends the substrate with a compact U(1) oriented sector, derives Maxwell-like dynamics in the weak-field limit, and proposes a candidate fixed-point formula for the fine-structure constant $\alpha_{\mathrm{EM}} \simeq 1/137$ based on the body-diagonal Wilson loop self-energy.

6. **Paper VII** — studies the joint Wilson-loop back-reaction of gravity and electromagnetism at the substrate scale, predicting a structural ratio $\alpha_G/\alpha_{\mathrm{EM}} = 1/\sqrt{2}$ and cross-coupling consequences observable in atomic clocks and cosmological $\dot{\alpha}/\alpha$.

The present paper, Paper I, supplies only the bounded conservative local rule and its constant-mobility Poisson baseline, validated by the $1/r$ verification of §6 and by the two-source superposition sanity check of §8.4. The support level and speculative content vary across the later papers and are stated explicitly in each.

References

[1] Konrad Zuse. *Rechnender Raum*. Vieweg, 1969.

[2] Edward Fredkin. Digital mechanics. *Physica D*, 45:254–270, 1990.

[3] Gerard 't Hooft. *The Cellular Automaton Interpretation of Quantum Mechanics*. Springer, 2016.

[4] Stephen Wolfram. *A New Kind of Science*. Wolfram Media, 2002.

[5] S. Lloyd. *Programming the Universe: A Quantum Computer Scientist Takes On the Cosmos*. Knopf, 2006.